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Building an Multi-agent System

The current **SeeWeeS Ops Reporting Agent** is a linear system: it retrieves data and documents, analyzes them, and generates a report in a single pass. However, real-world logistics for time-critical medicine are rarely linear. Data is often messy and incomplete; unpredictable weather, traffic, or resource limitations often disrupt the transportation plan; and high-stakes decisions require human oversight.

Your goal is to transform this prototype into a robust multi-agent system. You will extend the current LangGraph architecture to handle real-world operational friction, ensuring on-time deliveries from SeeWeeS. The starter repo now has a [data-for-enhancement](#) folder including artifacts for project ideas below.

Enhancement Ideas (Choose 1–2 as Your Core Focus)

Teams are encouraged to choose **one or two** of the following directions or propose their own high-impact business enhancement:

1. Self-Correction & Quality Assurance

Implement a cyclic “Audit Loop” where a specialized agent reviews the **PlannerAgent** output against the safety protocols and constraints extracted from the PDF. If the planner suggests a dispatch plan that violates a rule found in the PDF (e.g., “Must escalate if Risk Score is 3”), the system must loop back to the **PlannerAgent** with instructions to fix the error before generating the final HTML report.

2. “What-if” Scenario Simulation

Expand the system to allow users to specify hypothetical operational disruptions, such as a 20% demand spike, a warehouse closure, or driver shortage. The agents should simulate the impact on key performance indicators (KPIs) and generate contingency-based recommendations for each specific scenario.

3. Deep-Dive Trend Analysis

Enhance the **OpsDataAgent** to go beyond simple anomaly detection. You can implement Period-over-Period trend analysis or create logic that cross-references missing CSV data by searching the “Item Master Appendix” in the company playbook for potential product-class substitutions or identifier mappings.

4. Human-in-the-loop Workflows

Integrate a “Checkpoint” node in the LangGraph flow that pauses execution when high-risk conditions are met, such as a weather risk score exceeding a specific threshold. The system

should wait for a manager to review the current state and provide feedback or approval before the **ReportAgent** is allowed to finalize the plan.

5. Multi-Region or Multi-Day Resource Planning

Extend the current logic to handle complex planning across multiple geographic regions or extended time horizons. This involves comparing risks and KPIs across different delivery corridors and recommending how to allocate limited resources, like drivers, trucks, or temperature-controlled equipment, for maximum operational impact.

Project Tips & Guidance

To navigate this challenge, consider the following guiding principles as you develop your solution:

- Identify the “missing links” in your data

In professional analytics, you have to identify the relevant data to solve a complex problem. A core part of your objective is to identify which data points are missing from the current SeeWeeS data or documentation. You are expected to define these data, connect them with existing data, and, where needed, simulate realistic values as part of your exercise. This could include creating weather risk variables for the I-95 corridor, defining hospital priority levels, or adding resource planning constraints such as truck capacity. In addition, you are expected to define the KPIs that will guide your solution—clearly specifying how they are calculated, what data they depend on, and what thresholds indicate risk or success. This data augmentation and KPI definition together simulate the real-world process of data collection, problem framing, and decision-making..

- Integrate business complexity

A linear path is easy to build but lacks complexity in medical logistics. As you modify your LangGraph flow, ask yourself: “How does my system handle a failure?”. Our evaluation of your projects will favor teams that implement non-trivial logic—such as self-correction loops or “human-in-the-loop” checkpoints—that prevents the system from passing on bad information to leadership.

- Focus on “executive-ready” storytelling

Your model is only as useful as the decisions it enables. When designing your **ReportAgent** prompts or your final presentation, ensure the output is concise, structured, and decision-oriented. A high-quality report doesn’t just summarize data; it highlights the top risks, proposes concrete actions, and explains the “why” behind a recommendation so that a non-technical C-suite executive can act on it immediately.

- **Standardize your technical framework**

While creativity is encouraged, your solution must remain reproducible and well-organized. Ensure your repository includes a project background document that articulates your business problem, your data augmentation strategy, and your technical methodology. Clean code and a properly configured `.env.example` and `requirements.txt` are essential for showing that your system is ready for a production environment.

- **Ground your analytics in realistic constraints**

Your KPIs and risk metrics should be methodologically sound and aligned with the constraints of the SeeWeeS Operations Playbook. Ensure your agents are “grounded” in the business rules extracted from the documentation. We will look out for analytical results that surface meaningful patterns rather than just superficial summaries.

Deliverables

Teams must submit the following by **the deadline, Sunday May 10th, 2026 12:00PM PST** with the subject title, “[House Letters] Team # - UCLA MSBA AI Agents Project Challenge 2026”. **Email to the three instructors at yrong3249@g.ucla.edu, rohan.alternate@gmail.com, rohanmathur92@gmail.com, and CC Larry and the TAs.**

Your submission will be evaluated on completeness, clarity, and technical reproducibility.

1. GitHub Project Repository

Your repository should be a forked or organized copy of the starter repo and should include:

- **Source code:** All Python scripts (`graph.py`, `prompts.py`, `agents.py`, etc.) reflecting your architectural changes
- **Augmented dataset:** Any new CSV or PDF samples created to test your specific business scenario
- **Environment configuration:** A `requirements.txt` file and a `.env.example` template (ensure no secret keys are committed)
- **Deployment guide:** A `README.md` containing precise, step-by-step instructions for environment setup and a command-line example to run your end-to-end flow

2. Technical & Business Documentation

A comprehensive report (in pdf or md format) that bridges your technical work with business impact. It should cover but not limited to:

- **Executive summary:** Identify your specific stakeholder and the operational pain point you are solving

- **Key assumptions:** Detail your assumptions regarding logistics constraints, data availability, and business rules
- **Technical methodology:**
 - *Architectural enhancements:* Explain how you modified the LangGraph flow (e.g., new nodes, conditional edges, or cyclic loops)
 - *Agent design:* Describe any new tools or specialized prompts developed for your agents.
- **Results & validation:**
 - *Business insights:* Provide example outputs (screenshots or logs) and explain their significance for SeeWeeS operations
 - *Validation strategy:* Briefly mention how you verified the agent's logic and accuracy
- **Limitations & next steps:** Acknowledge any constraints in your current solution, and detail how you would scale this system with more time or real-world data access

Grades

- **Projects will be graded** based on the quality of the deliverables. Each team will receive either a rating of 3 (outstanding), 2 (average), or 1 (below average).
- **House points will be awarded** to members of the three best submissions. First place = 5 points per team member; Second place = 3 points per team member; Third place = 2 points per team member. Make sure to list your house name in the submission email subject title.
- **The first top team from each House will also deliver a 10-minute presentation** in Session 2 and will receive presentation points.